Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	AiX Global
Project / Solution Name	Clarity
Name (Leader)	Bhone Myat Paing Soe
Email & Phone (Leader)	jeremywin37@gmail.com ; 0173481780
University Affiliation(s)	Sunway University, Monash University, University of Nottingham Malaysia
Live Prototype URL	https://ai-x-global.replit.app/ Password: AixGlobal123456789
Code Repository URL (public / access granted to judges)	https://github.com/Mint3Ds/AiX-global

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The Clarity system utilizes a strictly business-scoped, multi-tier architecture to transform raw customer feedback into verifiable operational workflows. At the presentation layer, the user interface provides a partitioned workspace where synthetic identities manage AI-suggested themes, assign tasks, and review drafted public responses. This frontend communicates directly with a centralized backend API that enforces strict data isolation using a required business ID for all state queries and actions.

Once ingested, the backend orchestrates text processing through external AI APIs. The Jev model initially screens reviews for actionable feedback, utilizing OpenAI as a fallback. OpenAI then extracts exact-quote observations, maps them to dynamic operational themes, and later drafts contextual public replies based on resolved tasks. All end-to-end data provenance is securely persisted in a relational SQL database, immutably linking original review evidence, AI-extracted observations, and subsequent operational activity.

Finally, the system connects to external environments through specialized workflow adapters. Rather than immediately integrating with live cloud platforms, the current architecture locally simulates cloud event buses and external third-party APIs. This allows task handoffs and reply publications to be safely queued and verified without altering live systems. (Please refer to the appended System Block Diagram for visual details).

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript (v5.x)	Main type-safe programming language	Apache-2.0 License
Next.js (v14/15)	Full-stack React web framework & API routing	MIT License
React (v18/19)	Front-end UI component library	MIT License

Name & Version	Purpose in Project	License
PostgreSQL (v16)	Relational database management system	PostgreSQL License
Zod (v3.x)	Runtime schema and input validation	MIT License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
JEV	Classification of reviews into helpful / useless ; positive / negative ; etc.	API (paid)
OpenAI	Backup for classification ; basic analysis of review	API (paid)

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Live demonstration of prototype	Core (sponsor-provided)

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Jev 1.14.0 (latest)	Typesafe AI	Pre-trained	OpenAI service terms
GPT 5 Mini	OpenAI	Pre-trained	Typesafe service terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Harbour House Central	Self	Synthetic	N/a	N
Nusa Table Sunway	Self	Synthetic	N/a	N
Luma Salon	Self	Synthetic	N/a	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Codex with GPT-6.1 Sol	Skeleton of project, research, coding, front-end, graphic design	Major
Antigravity with Gemini 3.1 Pro and Gemini 4 Argon	Research, coding, front-end design, writing (ex.readme), proofreading, verification of requirements	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team treated AI-generated code and content as helpful starting points rather than final solutions. We validated AI outputs by integrating them into our codebase and running functional tests to ensure everything worked as intended. We reviewed the generated logic during our standard debugging and integration process to confirm it aligned with our project requirements. Any documentation or design suggestions were manually reviewed and adjusted for accuracy. Ultimately, our team takes full ownership of the final product, ensuring all AI contributions were actively tested and verified within our system.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Credits	Cloud deployment of live prototype and database management

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

While our Next.js and Drizzle ORM architecture is highly portable and could technically be migrated to alternative cloud providers, Replit was essential for delivering a fully functional prototype within the hackathon's strict timeframe. It provided the managed PostgreSQL database required for our multi-tenant data architecture and the secure environment secrets necessary to keep our external API keys strictly server-side. Without this sponsor resource, configuring a reliable, cloud-ready database and hosting environment would have significantly delayed our core application development.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No user data is collected.

Currently, the Clarity prototype operates exclusively on synthetic fixture data. No actual personal information is collected, stored, or transmitted to third-party AI services.

However, as we advance to integrate live feedback streams like Google Reviews, our production architecture will strictly comply with Malaysia's Personal Data Protection Act 2010 (PDPA, Act 709). The system will implement automated anonymization protocols to explicitly mask all Personally Identifiable Information (PII) before transmitting any live review payloads to external GenAI APIs (Jev and OpenAI) for screening and observation extraction. This ensures that while the AI analyzes the business context of a customer complaint, sensitive user identities remain securely isolated and protected within our Replit-managed database.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Clarity's AI components face risks including API rate limits, provider downtime, and potential model hallucinations. To mitigate dependency on a single provider, our architecture implements a reliable fallback mechanism: if the primary Jev classification model fails or is uncertain, the system automatically routes the request to an OpenAI model.
To address the critical risk of AI hallucination, where models might invent complaints, Clarity enforces exact-substring extraction. The AI must preserve the customer's original words as evidence rather than paraphrasing. Furthermore, we mitigate automated errors through strict human-in-the-loop safeguards. The system cannot resolve operational tasks automatically; it requires an explicit owner verification note. Additionally, if the AI detects unresolved complaints within a mixed review, it attaches a partial-reply warning to the draft. This warning intentionally restricts batch approval, ensuring mandatory human oversight before any external customer communication is finalized.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date] 10/25/2026